

Section 12.2: Vectors

Vectors and the Geometry of Space

MTH 310
Calculus III

Scalars vs. Vectors

In Calculus I and II, every quantity was a **number** — a single value on the number line.

Now we need quantities that carry both **size** and **direction**.

Scalars (magnitude only)	Vectors (magnitude AND direction)
Speed	Velocity
Distance	Displacement
Mass	Force
Length	Acceleration
Area	Weight (gravitational force)
Pressure	Work

A **scalar** has magnitude only. A **vector** has both magnitude AND direction.

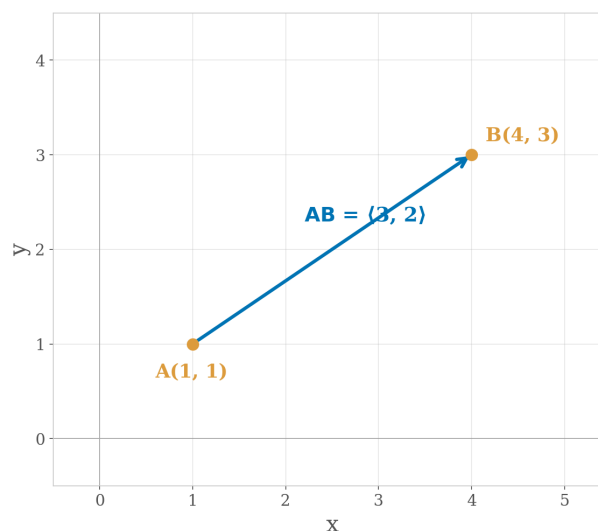
Displacement Vectors and Equal Vectors

A **displacement vector** $\vec{AB}$ represents moving from point A (initial point) to point B (terminal point).

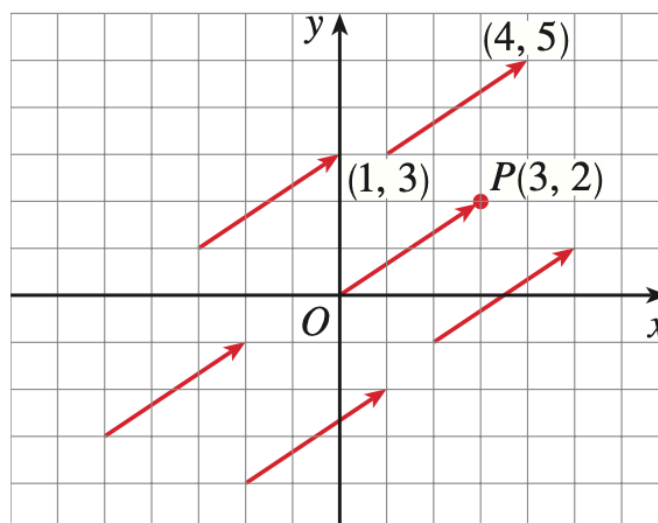
Equal Vectors

Two vectors are **equal** if they have the same magnitude and the same direction — even if they start at different points.

Equal vectors can be **translated** (slid) anywhere in space without changing the vector.



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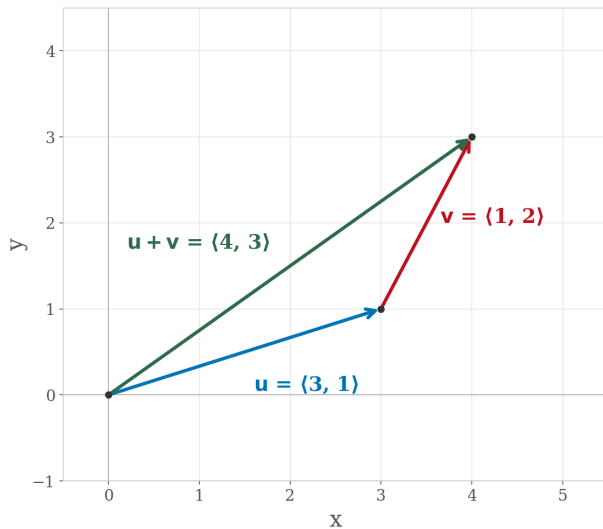


The **zero vector** $\vec{0}$ has magnitude zero and no specific direction. It is the only vector with this property.

Vector Addition

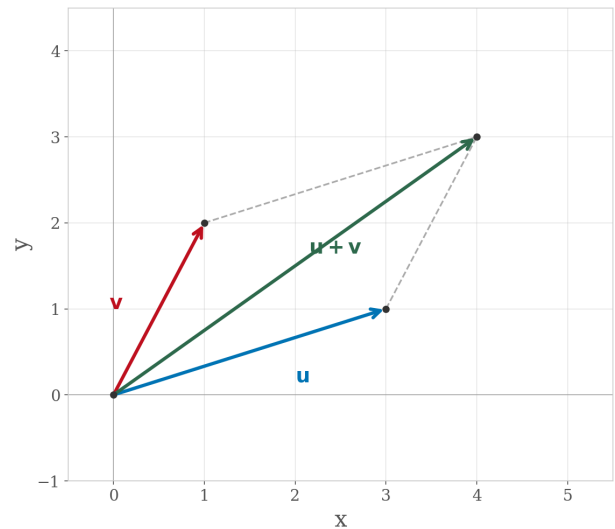
Two ways to add vectors geometrically:

Triangle Law: Place the tail of $\vec{v}$ at the tip of $\vec{u}$. The sum $\vec{u} + \vec{v}$ goes from the tail of $\vec{u}$ to the tip of $\vec{v}$.



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Parallelogram Law: Place the tails of $\vec{u}$ and $\vec{v}$ together. Complete the parallelogram. The diagonal is $\vec{u} + \vec{v}$.



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Chaining displacements: $\vec{AB} + \vec{BC} = \vec{AC}$

Walk from A to B , then from B to C . The net displacement is A to C .

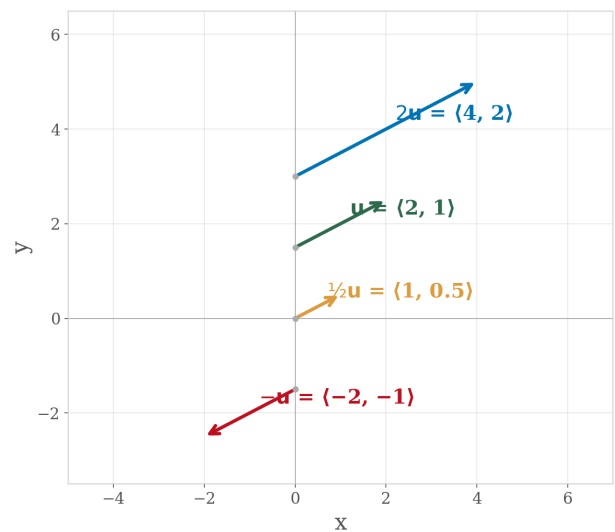
Vector addition is **commutative**: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$. Both the triangle and parallelogram laws give the same result.

Scalar Multiplication and Vector Difference

Scalar multiplication: $c\vec{u}$ stretches or shrinks $\vec{u}$ by a factor of $|c|$.

- If $c > 0$: same direction as $\vec{u}$
- If $c < 0$: opposite direction
- If $c = 0$: the zero vector

Parallel vectors: $\vec{u}$ and $\vec{v}$ are parallel if $\vec{v} = c\vec{u}$ for some scalar c .

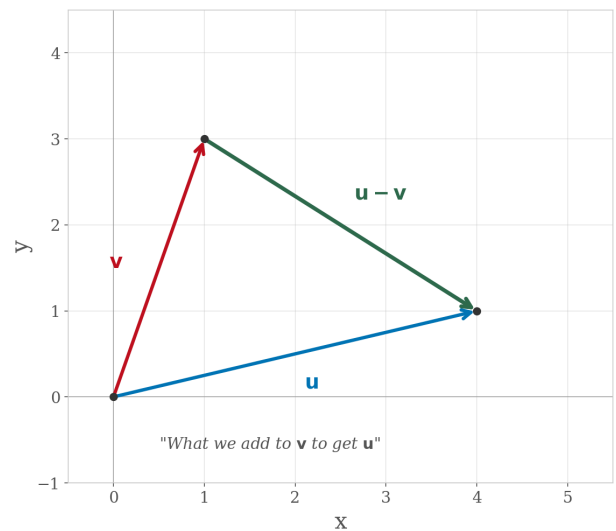


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Negative vector: $-\vec{u} = (-1)\vec{u}$ — same magnitude, opposite direction.

Vector difference: $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$

Think of $\vec{u} - \vec{v}$ as "what you add to $\vec{v}$ to get $\vec{u}$ " — like how $10 - 7.36 = 2.64$ answers "what do I add to 7.36 to get 10?"



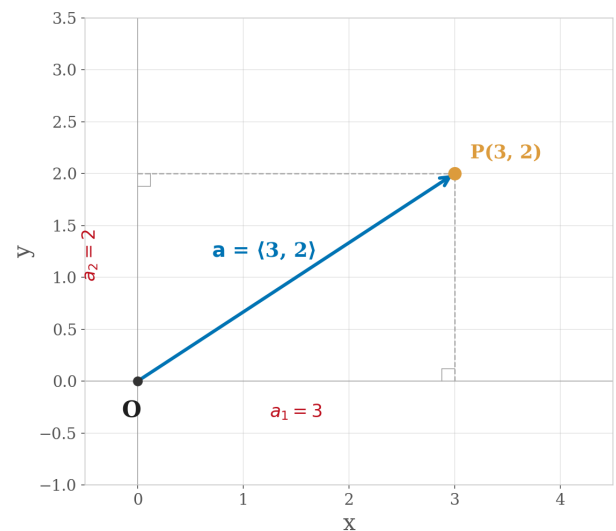
Components

If we place a vector with its initial point at the **origin**, it becomes a **position vector**. Its components are just its terminal coordinates.

In 2D: $\vec{a} = \langle a_1, a_2 \rangle$

In 3D: $\vec{a} = \langle a_1, a_2, a_3 \rangle$

We use **angle brackets** $\langle \cdot \rangle$ for vector components and **parentheses** $(\cdot)$ for points.



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Translating a vector to a different starting point gives an **equal** vector — same components, different location.

Displacement Vector Formula

Displacement Vector Formula

Given points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, the displacement vector from A to B is:

$$\vec{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Terminal minus initial.

Example 1

Find the displacement vector $\vec{AB}$ for $A(2, -3, 4)$ and $B(-2, 1, 1)$.

Subtract: terminal minus initial, component by component.

$$\vec{AB} = \langle -2 - 2, 1 - (-3), 1 - 4 \rangle = \langle -4, 4, -3 \rangle$$

Terminal minus initial. Always subtract in the order $B - A$ to get $\vec{AB}$.

Magnitude

Magnitude of a Vector

The **magnitude** (or length) of a vector $\vec{a} = \langle a_1, a_2, a_3 \rangle$ is:

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

In 2D: $|\vec{a}| = \sqrt{a_1^2 + a_2^2}$

This is the **distance formula** from Section 12.1! The magnitude of a position vector equals the distance from the origin to its terminal point.

Example: Vector Arithmetic

Example 2

For $\vec{a} = \langle 2, -1, 3 \rangle$ and $\vec{b} = \langle 0, 4, -3 \rangle$, compute:

1. $-3\vec{b}$
2. $|-3\vec{b}|$
3. $2\vec{a} + 5\vec{b}$

Compute each quantity step by step.

$$1. -3\vec{b} = -3\langle 0, 4, -3 \rangle = \langle 0, -12, 9 \rangle$$

$$2. |-3\vec{b}| = \sqrt{0^2 + (-12)^2 + 9^2} = \sqrt{0 + 144 + 81} = \sqrt{225} = 15$$

$$\text{Check: } |-3||\vec{b}| = 3\sqrt{0 + 16 + 9} = 3\sqrt{25} = 3 \cdot 5 = 15 \checkmark$$

$$3. 2\vec{a} + 5\vec{b} = 2\langle 2, -1, 3 \rangle + 5\langle 0, 4, -3 \rangle = \langle 4, -2, 6 \rangle + \langle 0, 20, -15 \rangle = \langle 4, 18, -9 \rangle$$

Property: $|c\vec{a}| = |c| |\vec{a}|$ — the magnitude of a scalar multiple equals the absolute value of the scalar times the magnitude.

Your Turn: Vector Arithmetic Practice

Example 3

Your turn! For $\vec{u} = \langle 3, -1, 2 \rangle$ and $\vec{v} = \langle -1, 4, 1 \rangle$, find:

1. $2\vec{u} - 3\vec{v}$
2. $|2\vec{u} - 3\vec{v}|$

Work through the scalar multiplications first, then subtract, then find the magnitude.

1. $2\vec{u} - 3\vec{v} = 2\langle 3, -1, 2 \rangle - 3\langle -1, 4, 1 \rangle = \langle 6, -2, 4 \rangle - \langle -3, 12, 3 \rangle = \langle 9, -14, 1 \rangle$
2. $|2\vec{u} - 3\vec{v}| = \sqrt{9^2 + (-14)^2 + 1^2} = \sqrt{81 + 196 + 1} = \sqrt{278}$

Standard Basis Vectors

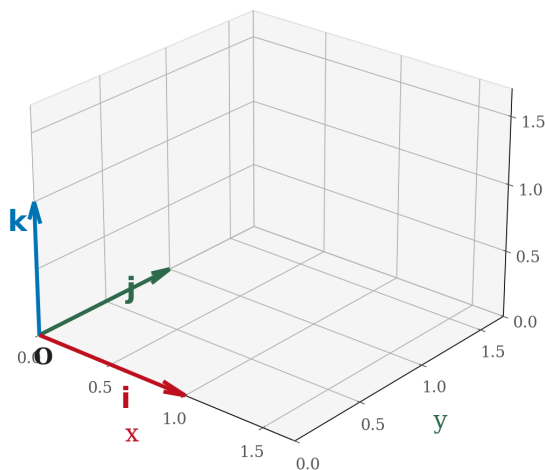
The **standard basis vectors** are unit vectors along each axis:

$$\vec{i} = \langle 1, 0, 0 \rangle, \quad \vec{j} = \langle 0, 1, 0 \rangle, \quad \vec{k} = \langle 0, 0, 1 \rangle$$

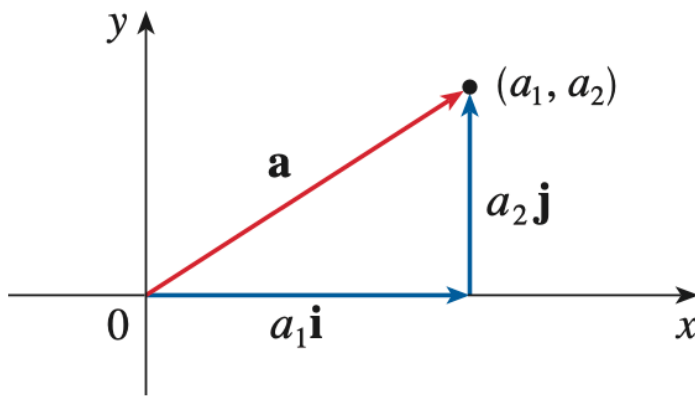
In 2D: $\vec{i} = \langle 1, 0 \rangle, \vec{j} = \langle 0, 1 \rangle$

Any vector can be written as a combination of basis vectors:

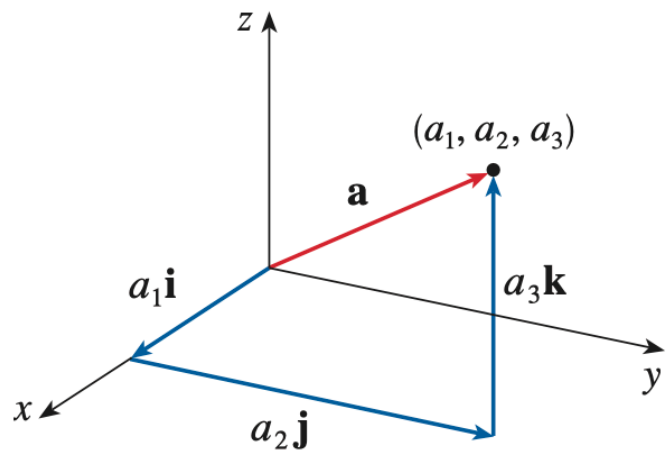
$$\vec{a} = \langle a_1, a_2, a_3 \rangle = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$



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(a) $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$



(b) $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Example: $\langle 3, 4 \rangle = 3\langle 1, 0 \rangle + 4\langle 0, 1 \rangle = 3\vec{i} + 4\vec{j}$

Unit Vectors

Unit Vector

A **unit vector** is a vector with magnitude 1.

To find the unit vector in the direction of $\vec{a}$:

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

This process is called **normalizing** the vector.

Example 4

Find the unit vector in the direction of $\vec{a} = \langle 3, 4 \rangle$.

Find $|\vec{a}|$, then divide each component by it.

$$|\vec{a}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\hat{a} = \frac{1}{5}\langle 3, 4 \rangle = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$$

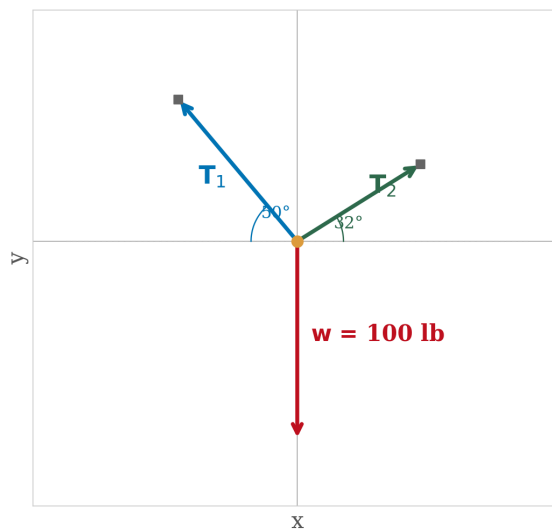
$$\text{Check: } \left| \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle \right| = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1 \checkmark$$

Divide by the magnitude to normalize. The unit vector $\hat{a}$ keeps the direction of $\vec{a}$ but has length 1.

Application: Forces in Equilibrium

Example 5

A 100-lb weight hangs from two wires as shown. Find the tensions $\vec{T}_1$ and $\vec{T}_2$ in the wires and the magnitudes of these tensions.



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Setting up the problem:

The system is in **equilibrium**, so the forces sum to zero:

$$\vec{T}_1 + \vec{T}_2 + \vec{w} = \vec{0}$$

Write each force in component form:

- $\vec{T}_1 = \langle -t_1 \cos 50^\circ, t_1 \sin 50^\circ \rangle$ \quad (pulls up-left)
- $\vec{T}_2 = \langle t_2 \cos 32^\circ, t_2 \sin 32^\circ \rangle$ \quad (pulls up-right)
- $\vec{w} = \langle 0, -100 \rangle$ \quad (gravity, straight down)

Solving the system:

Setting the x -components equal to zero:

$$-t_1 \cos 50^\circ + t_2 \cos 32^\circ = 0$$

Setting the y -components equal to zero:

$$t_1 \sin 50^\circ + t_2 \sin 32^\circ - 100 = 0$$

From the x -equation:

$$t_1 = \frac{t_2 \cos 32^\circ}{\cos 50^\circ}$$

Substitute into the y -equation:

$$\frac{t_2 \cos 32^\circ}{\cos 50^\circ} \sin 50^\circ + t_2 \sin 32^\circ = 100$$

$$t_2 (\cos 32^\circ \tan 50^\circ + \sin 32^\circ) = 100$$

$$t_2 = \frac{100}{\cos 32^\circ \tan 50^\circ + \sin 32^\circ} \approx 66.07 \text{ lb}$$

$$t_1 = \frac{66.07 \cos 32^\circ}{\cos 50^\circ} \approx 85.64 \text{ lb}$$

Homework

